$$= -\frac{\partial^2}{\partial p_x^2}\,\delta(p-p') \cdot p'$$

$$= p'\,\frac{\partial^2}{\partial p_x^2}\,\delta(p-p') \qquad = p'\left(\frac{i\partial}{\partial p_x}\right)^2 \delta(p-p')$$

$$\star\star \qquad \langle p|\hat{T}^2\hat{P}|p'\rangle = \langle p|\hat{X}^2\hat{P}^2|p\rangle p' = \langle p|\hat{X}^2|p\rangle p'$$

$$= -\frac{\partial^2}{\partial p_x^2}\,p\,\delta(p-p')$$

$$= \left(i\frac{\partial}{\partial p}\right)^2 p\,\delta(p-p')$$

$$\star \qquad \langle p|\hat{T}^2|p'\rangle = \langle p|\hat{X}^2\hat{P}^2|p'\rangle$$

(2) ذرة في صندوق : $\hat{T}^2 = \hat{X}^2\hat{P}^2$

$\hat{X} \to i\frac{\partial}{\partial p}$ $\qquad p \to p$

نقطة في صندوق ذو بعد واحد

$$= x\,x'\left(-i\frac{\partial}{\partial x}\,\delta(x-x')\right)$$

$$\star\star \qquad \langle x|\hat{X}^2\hat{P}^2|x'\rangle = x\,\langle x|\hat{P}^2|x'\rangle\,x'$$

$$\boxed{\text{ب}} \qquad (x\,p\,x)_{xx'} = x\left(-i\frac{\partial}{\partial x}\right)x\,\delta(x-x')$$

$$\star \qquad (x\,p\,x)_{xx'} = \langle x|\hat{X}\hat{P}\hat{X}|x'\rangle$$

(1)

حل المسألة $\Rightarrow$ $\hat{X} \to x$ $\qquad \hat{P} \to -i\frac{\partial}{\partial x}$ $\qquad$ في البعد الأول

$\hbar = \delta(x-x')$ نأخذ

(3) 证明：$F_{xx'} = -i\hbar^2 \frac{\partial}{\partial x}\delta(x-x')$

$$\hat{F}_{pp'} = \langle p|\hat{F}|p'\rangle$$

$$= \iint dx\,dx' \langle p|x\rangle\langle x|\hat{F}|x'\rangle\langle x'|p'\rangle$$

$$= \iint dx\,dx' \langle p|x\rangle F_{xx'}\langle x'|p'\rangle$$

$$= \iint dx\,dx' \langle p|x\rangle \left[-i\hbar^2\frac{\partial}{\partial x}\delta(x-x')\right]\langle x'|p'\rangle$$

$$\vdots$$

$$= -\frac{\partial^2}{\partial p^2}p'\delta(p-p')$$

$$= -\frac{\partial^2}{\partial p^2}p\,\delta(p-p')$$

$$\vdots$$

$$-\frac{\partial^2}{\partial p^2}p'\delta(p-p') = -p'\frac{\partial^2}{\partial p^2}\delta(p-p').$$

(3) 相互作用绘景 (Dirac 绘景)

设，$\hat{H}(t) = \hat{H}_0 + \hat{V}(t)$ （$\hat{H}_0$的解已知）

(a) 相互作用绘景态矢量

薛定谔绘景态矢

$$|\psi_I(t)\rangle \equiv \exp\left[\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]|\psi(t)\rangle$$

- 令 $t=t_0$；$|\psi_I(t_0)\rangle = |\psi(t_0)\rangle = |\psi_H(t_0)\rangle$

三个绘景重合！

$$\Rightarrow |\psi(t)\rangle = \exp\left[-\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]|\psi_I(t)\rangle$$

(b) 相互作用绘景算符

薛定谔绘景算符

$$\hat{A}_I(t) \equiv \exp\left[\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]\hat{A}\exp\left[-\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]$$

- 令 $t=t_0$：$\hat{A}_I(t_0) = \hat{A} = \hat{A}_H(t_0)$

三个绘景重合！

(c) $|\psi_I(t)\rangle$ 的时间演化

- $i\hbar\frac{d}{dt}|\psi_I(t)\rangle = \left[i\hbar\frac{d}{dt}\exp\left[\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]\right]|\psi(t)\rangle$

$$+ \exp\left[\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]\left(i\hbar\frac{d}{dt}|\psi(t)\rangle\right)$$

$$= -\hat{H}_0\exp\left[\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]|\psi(t)\rangle$$

$$+ \exp\left[\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]\hat{H}(t)|\psi(t)\rangle$$

$$= -\hat{H}_0|\psi_I(t)\rangle$$

$$+ \exp\left[\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]\left[\hat{H}_0+\hat{V}(t)\right]\exp\left[-\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]|\psi_I(t)\rangle$$

$$= \hat{V}_I(t)|\psi_I(t)\rangle$$

即

$$\boxed{i\hbar\frac{d}{dt}|\psi_I(t)\rangle = \hat{V}_I(t)|\psi_I(t)\rangle}$$

态的演化由相互作用 $\hat{V}$ 确定！

初始条件：

$$|\psi_I(t_0)\rangle = |\psi(t_0)\rangle = |\psi_H\rangle$$

(d) $\hat{A}_I(t)$ 对时间演化

$$\cdot\ i\hbar\frac{d}{dt}\hat{A}_I(t) = \left[i\hbar\frac{d}{dt}\exp\left[\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]\right]\hat{A}\exp\left[-\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]$$

$$+\exp\left[\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]\hat{A}\left[i\hbar\frac{d}{dt}\exp\left[-\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]\right]$$

$$= -\exp\left[\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]\hat{H}_0\hat{A}\exp\left[-\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]$$

$$+\exp\left[\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]\hat{A}\hat{H}_0\exp\left[-\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]$$

$$= \exp\left[\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]\left[\hat{A},\hat{H}_0\right]\exp\left[-\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]$$

$$= \left[\hat{A}_I(t),\hat{H}_0\right] \qquad (\hat{H}_{0I} = \hat{H}_0)$$

即

$$\boxed{i\hbar\frac{d}{dt}\hat{A}_I(t) = \left[\hat{A}_I(t),\hat{H}_0\right]}$$

算符的时间演化由 $\hat{H}_0$ 确定！

初始条件：

$$\hat{A}_I(t_0) = \hat{A} = \hat{A}_H(t_0)$$

(e) 경로적분 표현을 이용하여

$$\hat{A}_+ \equiv \langle \psi_F(t) | \hat{A}_I(t) | \psi_I(t) \rangle$$

$$= \langle \psi(t) | \hat{A}(t) | \psi(t) \rangle = \langle \psi_H(t) | \hat{A}_H(t) | \psi_H(t) \rangle$$

임을 보여라.

$$\hat{A}_+ = \left[i\hbar \frac{\delta}{\delta J} \langle \psi_I(t) | \right] \hat{A}(t) \left[-i\hbar \frac{\delta}{\delta J} | \psi_I(t) \rangle \right]$$

$$+ \langle \psi_I(t) | \hat{A}(t) \left[i\hbar \frac{\delta}{\delta J} | \psi_I(t) \rangle \right)$$

$$+ \langle \psi_I(t) | \left[i\hbar \frac{\delta}{\delta J} \hat{A}_I(t) \right] | \psi_I(t) \rangle$$

$$= \langle \psi(t) | \exp\left[-\frac{i}{\hbar} \hat{H}_0 (t-t_0) \right] \exp\left[\frac{i}{\hbar} \hat{H}_0 (t-t_0) \right] \hat{A}$$

$$\times \exp\left[-\frac{i}{\hbar} \hat{H}_0 (t-t_0) \right] \exp\left[\frac{i}{\hbar} \hat{H}_0 (t-t_0) \right] | \psi(t) \rangle$$

(e) 경로적분 표현을 이용하여

$$= -\langle \psi_I(t) | \hat{V}_I(t) \hat{A}_I(t) | \psi_I(t) \rangle$$

$$+ \langle \psi_I(t) | [\hat{A}_I(t), \hat{H}_0] | \psi_I(t) \rangle$$

$$+ \langle \psi_I(t) | \hat{A}_I(t) \hat{V}_I(t) | \psi_I(t) \rangle$$

$$= \langle \psi_I(t) | [\hat{A}_I(t), \hat{V}_I(t)] | \psi_I(t) \rangle$$

$$+ \langle \psi_I(t) | [\hat{A}_I(t), \hat{H}_0] | \psi_I(t) \rangle$$

$$= \langle \psi_I(t) | [\hat{A}_I(t), \hat{H}_I(t)] | \psi_I(t) \rangle .$$

考虑极限： $[\hat{A}_I(t), \hat{H}_I(t)] \to 0$

$$\exp\left[\frac{i}{\hbar}\hat{H}_0(t-t_0)\right] [\hat{A}, \hat{H}(t)] \exp\left(-\frac{i}{\hbar}\hat{H}_0(t-t_0)\right) \to 0$$

$$\Rightarrow \quad [\hat{A}, \hat{H}(t)] \to 0 .$$

(4) 相互作用表象中的时间演化算符 $\hat{U}_I(t,t_0)$

定义式：

$$|\psi_I(t)\rangle \equiv \hat{U}_I(t,t_0)|\psi_I(t_0)\rangle$$

- $\hat{U}_I(t,t) = \mathbb{1}$
- $\hat{U}_I^\dagger(t,t_0) = \hat{U}_I^{-1}(t,t_0)$ （幺正性）
- $\hat{U}_I(t_2,t_0) = \hat{U}_I(t_2,t_1)\hat{U}_I(t_1,t_0)$ （传递性）
- $\hat{U}_I^{-1}(t,t_0) = \hat{U}_I(t_0,t)$.

时间演化：

$$i\hbar\frac{d}{dt}\hat{U}_I(t,t_0) = \hat{V}_I(t)\hat{U}_I(t,t_0)$$

（初始条件：$\hat{U}_I(t_0,t_0) = \mathbb{1}$）

. $\hat{U}_I(t,t_0)$ 的幂级数表达式

$$\hat{U}_I(t,t_0) = \mathbb{1} + \frac{1}{i\hbar}\int_{t_0}^{t} dt_1\, \hat{V}_I(t_1)$$

零阶 t_0 一阶

$$+ \frac{1}{(i\hbar)^2}\int_{t_0}^{t} dt_1 \int_{t_0}^{t_1} dt_2\, \hat{V}_I(t_1)\hat{V}_I(t_2)$$

二阶

$$+ \cdots \quad \frac{1}{(i\hbar)^n} \text{n阶}$$

$$\hat{U}_I(t,t_0) = \hat{T}\exp\left[\frac{1}{i\hbar}\int_{t_0}^{t} dt'\, \hat{V}_I(t')\right]$$

(8) $\hat{U}(t,t_0)$ 与 $\hat{U}_I(t,t_0)$ 的关系

$$\hat{U}_I(t,t') = e^{\frac{i\hat{H}_0}{\hbar}(t-t_0)}\hat{U}(t,t')e^{-\frac{i\hat{H}_0}{\hbar}(t'-t_0)}$$

(证明略)

· $|\psi_I(t)\rangle = \sum_i |a_i\, t\rangle_I\, {}_I\langle a_i\, t|\psi_I(t)\rangle$

* 海森伯绘景中，态与基矢都随时间变化！

· 例：对外界微扰的线性响应.

(h) A - 表象

· $\hat{A}_I(t) = \exp\left[\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]\hat{A}\exp\left[-\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]$.

· $|a_i\,t\rangle_I = \exp\left[\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]|a_i\rangle$

其中. $\boxed{\hat{A}|a_i\rangle = a_i|a_i\rangle}$

· $\hat{A}_I(t)\,|a_i\,t\rangle_I$

$= \exp\left[\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]\hat{A}\exp\left[-\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]\exp\left[\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]|a_i\rangle$

$= \exp\left[\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]\hat{A}|a_i\rangle = a_i\exp\left[\frac{i}{\hbar}\hat{H}_0(t-t_0)\right]|a_i\rangle$

$= a_i|a_i\,t\rangle_I$

即 $\boxed{A_I(t)\,|a_i\,t\rangle_I = a_i\,|a_i\,t\rangle_I}$

·
$$\begin{cases} {}_I\langle a_i\,t|\,a_j\,t\rangle_I = \delta_{ij} & \text{(正交性)} \\ \sum_i |a_i\,t\rangle_I\,{}_I\langle a_i\,t| = \mathbb{1} & \text{(完备性)} \end{cases}$$

基矢: $\{|a_i\,t\rangle_I\}$

相互作用绘景下的

A - 表象

设. $\hat{H}(t) = \hat{H}_0 + \theta(t-t_0)\hat{V}(t)$ → 外界微扰

. $\overline{\hat{A}(t)} = \langle \psi_I(t) | \hat{A}_I(t) | \psi_I(t) \rangle$.

• $|\psi_I(t)\rangle = \hat{U}_I(t,t_0)|\psi_I(t_0)\rangle$

$= \hat{T}\exp\left[-\frac{i}{\hbar}\int_{t_0}^{t} dt' \hat{V}_I(t')\right]|\psi_I(t_0)\rangle$

$\approx \left(1 - \frac{i}{\hbar}\int_{t_0}^{t} dt' \hat{V}_I(t')\right)|\psi_I(t_0)\rangle$

$\Rightarrow \overline{\hat{A}(t)} \approx \langle \psi_I(t_0)| \left(1 + \frac{i}{\hbar}\int_{t_0}^{t} dt' \hat{V}_I(t')\right) \hat{A}_I(t)$

$\left(1 - \frac{i}{\hbar}\int_{t_0}^{t} dt'' V_I(t'')\right)|\psi_I(t_0)\rangle$

$\approx \langle \psi_I(t_0)|\hat{A}_I(t)|\psi_I(t_0)\rangle$

零阶

$+ \frac{i}{\hbar}\langle \psi_I(t_0)|\int_{t_0}^{t} dt' \left[\hat{V}_I(t'), \hat{A}_I(t)\right]|\psi_I(t_0)\rangle$.

一阶 (线性响应)

§2.4 量子力学的路径积分形式

量子力学的三种表述方式

(1) 薛定谔的波动力学

(2) 海森伯的矩阵力学

(3) 费曼的路径积分形式

(1) 传播子的定义

- $|\psi(t)\rangle = \hat{U}(t,t_0)\,|\psi(t_0)\rangle$

时间演化算符

$$\Rightarrow \quad \langle x|\psi(t)\rangle = \langle x|\hat{U}(t,t_0)|\psi(t_0)\rangle$$

$$= \int_{-\infty}^{\infty} dx' \,\langle x|\hat{U}(t,t_0)|x'\rangle\langle x'|\psi(t_0)\rangle$$

$$= \int_{-\infty}^{\infty} dx' \, K(xt;x't_0)\langle x'|\psi(t_0)\rangle .$$

$$\boxed{K(xt;x't_0) \equiv \langle x|\hat{U}(t,t_0)|x'\rangle}$$

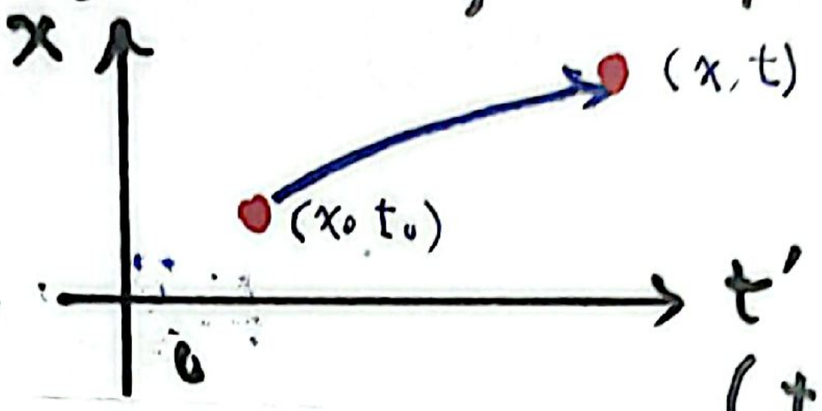
时间演化算符在坐标表象中的矩阵元！

- $\psi(x,t) \equiv \langle x|\psi(t)\rangle$

$$\Rightarrow \quad \boxed{\psi(x,t) = \int_{-\infty}^{\infty} dx' \, K(xt;x't_0)\,\psi(x',t_0)}$$

4 传播子概念

• 设初始 t_0 时刻粒子位于 x_0 处，则称 $t>t_0$ 时刻在 x 处发现粒子的概率幅为传播子。



$(t>t_0)$

• 初始态：

$\hat{x}|x_0\rangle = x_0|x_0\rangle$，则称 $|x_0\rangle$ 为 $\hat{x}$ 的本征态，本征值为 x_0。

波函数：$\psi(x,t) = \langle x|x_0\rangle$

$= \delta(x-x_0)$

• $$\psi(x,t) = \int_{-\infty}^{\infty} dx' K(xt;x't_0)\,\delta(x'-x_0)$$

$$= K(xt;x_0t_0) \quad (\text{即为传播子！})$$

(2) $K(xt;x_0t_0)$ 的时间演化

$$i\hbar\frac{\partial}{\partial t}K(xt;x_0t_0) = i\hbar\frac{\partial}{\partial t}\left[\langle x|\hat{U}(t,t_0)|x_0\rangle\right]$$

$$= \langle x|\left[i\hbar\frac{\partial}{\partial t}\hat{U}(t,t_0)\right]|x_0\rangle = \langle x|\hat{H}(t)\hat{U}(t,t_0)|x_0\rangle$$

$$= \int_{-\infty}^{\infty} dx' \langle x|\hat{H}(t)|x'\rangle\langle x'|\hat{U}(t,t_0)|x_0\rangle$$

$\hat{H}(\hat{x}, \hat{p}, t)$

$$= \int_{-\infty}^{\infty} dx' \, H_{xx'}(t) \, \langle x' | \hat{U}(t, t_0) | x_0 \rangle$$

$$= \int_{-\infty}^{\infty} dx' \left[H\left(x, -i\hbar \frac{\partial}{\partial x}, t\right) \delta(x - x') \right] \langle x' | \hat{U}(t, t_0) | x_0 \rangle$$

$$= H\left(x, -i\hbar \frac{\partial}{\partial x}, t\right) \int_{-\infty}^{\infty} dx' \, \delta(x - x') \langle x' | \hat{U}(t, t_0) | x_0 \rangle$$

$$= H\left(x, -i\hbar \frac{\partial}{\partial x}, t\right) \langle x | \hat{U}(t, t_0) | x_0 \rangle$$

$$= H\left(x, -i\hbar \frac{\partial}{\partial x}, t\right) K(xt ; x_0 t_0)$$

故

$$i\hbar \frac{d}{dt} K(xt ; x_0 t_0) = H\left(x, -i\hbar \frac{\partial}{\partial x}, t\right) \times K(xt ; x_0 t_0)$$

初始条件：

$$K(x t_0 ; x_0 t_0) = \delta(x - x_0)$$

下证明：

$$K(xt;x_0t_0)=\langle x|\hat{U}(t,t_0)|x_0\rangle$$

令 $t=t_0$，

$$K(xt_0;x_0t_0)=\langle x|\hat{U}(t_0,t_0)|x_0\rangle$$

$$=\langle x|x_0\rangle=\delta(x-x_0).$$

证毕.

※ 若 $\hat{H}$ 不显含时间，则

$$\hat{U}(t,t_0)=e^{-\frac{i}{\hbar}\hat{H}(t-t_0)}.$$

$$K(xt;x_0t_0)=\langle x|e^{-\frac{i}{\hbar}\hat{H}(t-t_0)}|x_0\rangle$$

$$=\langle x|e^{-\frac{i}{\hbar}\hat{H}t}\quad e^{\frac{i}{\hbar}\hat{H}t_0}|x_0\rangle$$

$$=\langle x,t|x_0,t_0\rangle_H,\quad \text{内积}$$

$$\hat{X}_H(t)\,|x,t\rangle_H = x\,|x,t\rangle_H$$

$$\begin{cases} {}_H\langle x',t|\,x,t\rangle_H = \delta(x'-x) \\ \int dx\ |x,t\rangle_H\ {}_H\langle x,t| = \mathbb{1} \end{cases}$$

(3) $K(xt;x_0t_0)$ の満たす性質（組合規則）

$$K(xt;x_0t_0) = \langle x|\hat{U}(t,t_0)|x_0\rangle$$

$$= \langle x|\hat{U}(t,t_1)\hat{U}(t_1,t_0)|x_0\rangle$$

$$= \int_{-\infty}^{\infty} dx_1\ \langle x|\hat{U}(t,t_1)|x_1\rangle\langle x_1|\hat{U}(t_1,t_0)|x_0\rangle$$

$$= \int_{-\infty}^{\infty} dx_1\ K(xt;x_1t_1)\,K(x_1t_1;x_0t_0)$$

$$= \int \left(\prod_{j=1}^{N-1} dx_j\right) \prod_{i=0}^{N-1} K(x_{i+1}t_{i+1};x_it_i) \quad \begin{cases} x_N = x \\ t_N = t \end{cases}$$

（将 $t-t_0$ 分成 N 个等份）

传播子的计算（能量表象）

（$\hat{H}$ 不显含时间：$\hat{U}(t,t_0) = e^{-\frac{i}{\hbar}\hat{H}(t-t_0)}$）

$$\begin{cases} \hat{H}\,|E_n\rangle = E_n |E_n\rangle \\ \langle E_{n'}|E_n\rangle = \delta_{n'n} \\ \sum_n |E_n\rangle\langle E_n| = \mathbb{1} \end{cases} \cdot \begin{cases} H\left(x, -i\hbar\frac{\partial}{\partial x}\right)\psi_n(x) \\ = E_n \psi_n(x) \\ \int dx\, \psi_{n'}^*(x)\,\psi_n(x) = \delta_{n'n} \\ \sum_n \psi_n^*(x')\,\psi_n(x) = \delta(x-x') \end{cases}$$

$$\cdot\quad K(x\,t\,;x_0 t_0) = \langle x|\, e^{-\frac{i}{\hbar}\hat{H}(t-t_0)}\,|x_0\rangle$$

$$= \sum_n \langle x|\, e^{-\frac{i}{\hbar}\hat{H}(t-t_0)}\,|E_n\rangle\langle E_n|\,x_0\rangle$$

$$= \sum_n e^{-\frac{i}{\hbar}E_n(t-t_0)}\,\langle x|E_n\rangle\langle E_n|x_0\rangle \quad \Rightarrow$$

$$\boxed{K(x\,t\,;x_0 t_0) = \sum_n e^{-\frac{i}{\hbar}E_n(t-t_0)}\,\psi_n(x)\,\psi_n^*(x_0)}$$

· 薛定谔方程的解？　　验证 ⇒

(左)：记录 $K(xt;x_0t_0)$

$$= i\hbar\frac{\partial}{\partial t}\left[\sum_n e^{-\frac{i}{\hbar}E_n(t-t_0)}\psi_n(x)\psi_n^*(x_0)\right]$$

$$= \sum_n E_n e^{-\frac{i}{\hbar}E_n(t-t_0)}\psi_n(x)\psi_n^*(x_0)$$

(右)：$H(x,-i\hbar\frac{\partial}{\partial x})K(xt;x_0t_0)$

$$= H(x,-i\hbar\frac{\partial}{\partial x})\left[\sum_n e^{-\frac{i}{\hbar}E_n(t-t_0)}\psi_n(x)\psi_n^*(x_0)\right]$$

$$= \sum_n E_n e^{-\frac{i}{\hbar}E_n(t-t_0)}\psi_n(x)\psi_n^*(x_0) \quad (ok)$$

· 初始条件？　　验证 ⇒

$$K(xt_0;x_0t_0) = \sum_n \psi_n(x)\psi_n^*(x_0)$$

$$= \sum_n \langle x|E_n\rangle\langle E_n|x_0\rangle = \langle x|x_0\rangle = \delta(x-x_0)$$

(ok)

· 令 $x = x_0$,

$$K(xt; xt_0) = \sum_n e^{-\frac{i}{\hbar} E_n (t - t_0)} |\psi_n(x)|^2 \quad \cdots (A)$$

两边积分，

$$\int_{-\infty}^{\infty} dx\, K(xt; xt_0) = \sum_n e^{-\frac{i}{\hbar} E_n (t - t_0)} \int_{-\infty}^{\infty} dx\, |\psi_n(x)|^2$$

$$= \sum_n e^{-\frac{i}{\hbar} E_n (t - t_0)} \quad \cdots (B).$$

例 1. 一维自由粒子

$$\hat{H} = \frac{\hat{p}^2}{2m} ,$$

$$|E_n\rangle \to |p\rangle ,$$

$$E_n \to \frac{p^2}{2m} ,$$

$$\sum_n |E_n\rangle\langle E_n| = \mathbb{1} \longrightarrow \int dp\, |p\rangle\langle p| = \mathbb{1}$$

$$\langle x|E_n\rangle \rightarrow \langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} \exp\left(\frac{i}{\hbar} p x\right) .$$

- $$K(xt;x_0t_0) = \langle x| e^{-\frac{i\hat{H}(t-t_0)}{\hbar}} |x_0\rangle$$

$$= \langle x| e^{-\frac{i}{\hbar}\frac{\hat{p}^2}{2m}(t-t_0)} |x_0\rangle$$

$$= \int_{-\infty}^{\infty} dp \langle x| e^{-\frac{i}{\hbar}\frac{\hat{p}^2}{2m}(t-t_0)} |p\rangle\langle p|x_0\rangle$$

$$= \int_{-\infty}^{\infty} dp \langle x| e^{-\frac{i}{\hbar}\frac{p^2}{2m}(t-t_0)} |p\rangle\langle p|x_0\rangle$$

$$= \int_{-\infty}^{\infty} dp \; e^{-\frac{i}{\hbar}\frac{p^2}{2m}(t-t_0)} \frac{1}{\sqrt{2\pi\hbar}} e^{\frac{i}{\hbar}px} \frac{1}{\sqrt{2\pi\hbar}} e^{-\frac{i}{\hbar}px_0}$$

$$= \frac{1}{2\pi\hbar}\int_{-\infty}^{\infty} dp \; \exp\left[\frac{i}{\hbar}\left((x-x_0)p - \frac{p^2(t-t_0)}{2m}\right)\right]$$

利用 Fresnel 积分公式

$$\int_{-\infty}^{\infty} dy\ \exp\left[-i\left(ay^2+by\right)\right]=\sqrt{\frac{\pi}{ia}}\ e^{ib^2/4a}$$

$$=\sqrt{\frac{m}{i2\pi\hbar(t-t_0)}}\ e^{\frac{im(x-x_0)^2}{2\hbar(t-t_0)}}$$

即

$$K(xt;x_0t_0)=\sqrt{\frac{m}{i2\pi\hbar(t-t_0)}}\ e^{\frac{im(x-x_0)^2}{2\hbar(t-t_0)}}$$

例2. 一维线性谐振子

$$\hat{H}=\frac{\hat{p}^2}{2m}+\frac{m\omega^2\hat{x}^2}{2},\quad E_n=\left(n+\frac{1}{2}\right)\hbar\omega.$$

$(n=0,1,2,\cdots)$

$$\psi_n(x)=\left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}}\left(2^n n!\right)^{-\frac{1}{2}}H_n\left(\sqrt{\frac{m\omega}{\hbar}}x\right)e^{-\frac{m\omega}{2\hbar}x^2}.$$

$$K(xt;x_0t_0) = \sum_n e^{-\frac{iE_n}{\hbar}(t-t_0)} \psi_n(x)\,\psi_n^*(x_0) \quad .$$

利用 Mehler 公式：

$$\exp\left[-(x^2+y^2)\right] \sum_{n=0}^{\infty} \left(\frac{\xi^n}{2^n n!}\right) H_n(x)\,H_n(y)$$

$$= \frac{1}{\sqrt{1-\xi^2}} \exp\left[-\frac{(x^2+y^2-2xy\xi)}{(1-\xi^2)}\right]$$

$$\Rightarrow$$

$$k(xt;x_0t_0) = \sqrt{\frac{m\omega}{i2\pi\hbar \sin\omega(t-t_0)}}$$

$$\exp\left[\frac{i}{\hbar}\frac{m\omega}{2\sin\omega(t-t_0)}\left[(x^2+x_0^2)\cos\omega(t-t_0) - 2xx_0\right]\right]$$

(a) 当 $\omega \to 0$ 时，得到自由粒子的传播子

(b) 初始条件

利用 δ- 函数的表达式

$$\frac{1}{\sqrt{\pi}} \lim_{t \to 0} \left\{ \frac{1}{\sqrt{t}} \exp\left[-\frac{(x-x_0)^2}{t} \right] \right\} = \delta(x-x_0)$$

即有

$$K(x t_0 ; x_0 t_0) = \delta(x-x_0)$$

(c) 由 $K \to \psi_n , E_n$